Statistical Machine Translation System Combinations on Phrase-based, Hierarchical Phrase-based and Operation Sequence Model for Burmese and Pa'O Language Pair

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myMediCon: End-to-End Burmese Automatic Speech Recognition for Medical Conversations

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Language Understanding λ(μ) Laboratory

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Evaluating PBSMT, HPBSMT, and OSM Approaches

Research comparing Phrase-Based SMT, Hierarchical Phrase-Based SMT, and Operation Sequence Model to optimize translation quality between Burmese and Pa'O languages.

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Introduction

Research Motivation

LINGUISTIC DIVERSITY

Myanmar's Rich Linguistic Landscape

Burmese serves as the official language, while Pa'O spoken by the Pa'O people remains significantly underrepresented in digital and NLP resources.

THE COMMUNICATION GAP

A Critical Communication Barrier

Many Pa'O speakers have limited proficiency in Burmese, and vice versa. This language gap restricts access to education, public services, and information for ethnic minority communities.

THE RESEARCH NEED

Machine Translation as a Bridge

Developing an effective Burmese-Pa'O machine translation system can empower communities, preserve cultural heritage, and advance NLP research for low-resource Southeast Asian languages.

"This paper presents the first systematic evaluation of SMT system combinations for the Burmese-Pa'O language pair."

The Language Challenge

Bridging Communication Gaps Between Burmese and Pa'O Languages

Myanmar is home to about 100 languages. Pa'O spoken by roughly 500,000 people across Shan, Kayah, Mon, Bago and parts of northern Thailand - has limited access to translation resources. This research develops machine translation tools between Burmese and Pa'O to improve communication and information access for minority communities.

~100	500,000	5
Languages Spoken in Myanmar	Pa'O Language Speakers	Major Regions of Pa'O Speakers

LINGUISTIC FOUNDATIONS

Understanding Pa'O: A Central Karenic Language with Unique Phonetic Features

Pa'O is a Central Karenic language belonging to the Tibeto-Burman language family, sharing the same family as Burmese. Although Pa'O uses the Burmese script and alphabet, it possesses distinct phonetic characteristics and pronunciation patterns.

- **Subject-Verb-Object (SVO) sentence structure**
- **Words written with multiple syllables**
- **Unique system of phonetics with special characters**

Key Linguistic Features

33 Consonants

8 Independent Vowels

16 Dependent Vowels

3 Medial Diacritics

Language Background

Characteristics of the Pa'O Language

Pa'O is a Tibeto-Burman minority language spoken by over half a million people, primarily in Shan State, Myanmar.

Tibeto-Burman Roots

Pa'O belongs to the Tibeto-Burman language family, one of Southeast Asia's oldest and most diverse language groups, sharing deep structural roots with Burmese.

SVO Word Order

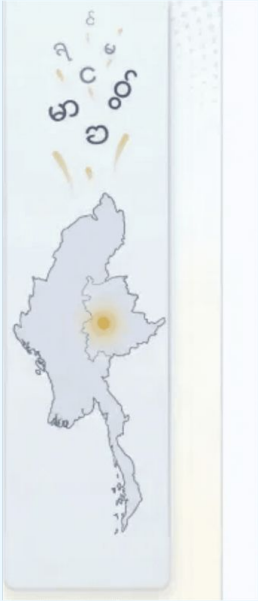
Pa'O follows Subject-Verb-Object sentence structure, notably different from Burmese's SOV order. This structural divergence poses key challenges for direct machine translation.

Script Similarities

Pa'O uses the Burmese script as its writing system, sharing the same alphabet and Unicode encoding, yet the pronunciation and phonological mapping differ significantly.

Unique Phonetic Markers

Pa'O employs distinct tonal features and syllable segmentation rules. Syllables are written as compound units, requiring specialized tokenization strategies for NLP processing.



LANGUAGE EXAMPLES

Parallel Sentences: Burmese and Pa'O

bm: သူ သစ်ပင် တွေ ပန်းပင် တွေ စိုက်တယ် ။
po: ဝွေ့ ဆို့ သေင်းမ္ပူ ဖုံ ကဝ်းမ္ပူ ဖုံ ။
English: He planted trees and flowers.

bm: မင်း အကြိုက်ဆုံး ရာသီဥတု က ဘာလဲ ။
po: နာ့ အကျိုက်သွတ် ရားသီဥတု နဝ့် တမ့ဲင့်ဟောင်း ။
English: What is your favorite weather?

bm: ခင်ဗျား ပြောခဲ့ သလို ကျွန်တော် ရှင်းပြ ခဲ့တယ် ။
po: နာ့ ကဒေါ့ အတ္ပိုင့် ခွေ သျင့်ပျ ဗားဒျား ။
English: I explained as you said.

Examples of parallel sentences showing the translation alignment between Burmese (bm), Pa'O (po), and English reference.

RESEARCH METHODOLOGY

Comparing Three SMT Approaches for Optimal Translation Performance

Phrase-Based SMT (PBSMT)

Operates on phrasal units and uses noisy channel models to maximize translation probability based on source sentences. The model translates a Burmese sentence into a Pa'O sentence using the equation:

```
t = argmax_t P(t|s) = argmax_t P(s|t)P(t)
```

Hierarchical Phrase-Based SMT (HPBSMT)

Extends classical PBSMT by incorporating rule expressivity through synchronous context-free grammar (SCFG) rules. These rules contain non-terminals that can be expanded, enabling sophisticated translation patterns for language pairs requiring long-distance reordering.

Operation Sequence Model (OSM)

Represents an N-gram-based translation and reordering model that treats aligned bilingual corpus as a sequence of operations and learns a Markov model. Operation types include generate (creation of words), insert gap (explicit target positions), and jump (reordering).

HPBSMT MODEL

Hierarchical Phrase-Based Grammar Rules

[X][X] ဘယ်သူ လဲ [X] ||| [X][X] ပါ့မ့ဲ့ [X]
[X][X] ဘယ်သူ လဲ [X] ||| [X][X] ပါ့မ့ဲ့ ဖုံးဟောင်း [X]
[X][X] ဘယ်သူ လဲ [X] ||| [X][X] ပါ့မ့ဲ့ ဟောင်း [X]
[X][X] ဘယ်သူ လဲ [X] ||| [X][X] ပါ့မ့ဲ့ အီ [X]
[X][X] ဘယ်သူ လဲ [X] ||| [X][X] ပါ့မ့ဲ့ဟောင်း [X]

An example of hierarchical phrase-based grammar rules from an HPBSMT model, demonstrating the alignment between Burmese and Pa'O using SCFG grammar.

OSM Translation Process: Practical Example

Source: ဝွေ့မူး ကျိုက်ဒျား ကဝ်းရား
Target: သူမ ပန်းသီး ကြိုက်တယ်

Operation 1: Generate (ဝွေ့မူး, သူမ)
Operation 2: Insert Gap
Operation 3: Generate (ကဝ်းရား, ပန်းသီး)
Operation 4: Jump Back (1)
Operation 5: Generate (ကျိုက်ဒျား, ကြိုက်တယ်)

A step-by-step demonstration of the Operation Sequence Model (OSM) translation process from a Pa'O sentence to Burmese.

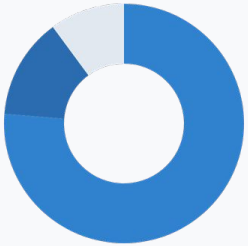
EXPERIMENTAL SETUP

Building the Foundation: Corpus Preparation and Segmentation Strategies

ASEAN-MT Parallel Corpus

Utilized 18,354 Burmese sentences (without name entity tags) in the travel domain, comprising six main categories: greeting/communication, survival, food, fun, resource, and special needs.

Data Split



- Training (14,000)
- Development (2,500)
- Evaluation (1,854)

Segmentation Schemes

- **Word Segmentation:** Manual word segmentation rules for Pa'O.
- **Syllable Segmentation:** RE-based syllable segmentation tool ("sylbreak").
- **Syllable-SentencePiece:** Language-independent subword tokenizer and detokenizer.

https://github.com/ye-kyaw-thu/sylbreak

Tools Employed

Moses Toolkit
PBSMT, HPBSMT, and OSM training

GIZA++
Word alignment

KenLM
5-gram language model training

MERT
Decoder parameter tuning

Segmentation Strategies for Pa'O Sentences

Word Segmentation

Separates sentence into individual words with meaning mappings.

Unsegmented sentence:
po: ဝွေ့မူးခြင်းနှယ်လိုပေးဖုံးဝဝ်းခြင်းတပှဒ်။

Segmented sentence:
po: ဝွေ့မူး (“She”) ခြင်းနှယ် (“tells”) လိုပေး ဖုံး (“children”) ဝဝ်းခြင်း တ ပှဒ် (“a story”) ။ (“.”)

Syllable Segmentation

Breaks down words into their constituent syllables.

Syllable segmented Pa’O sentence:
po: ဝွေ့ မူး ခြင်း နှယ် လို ပေး ဖုံး ဝဝ်း ခြင်း တ ပှဒ်။

SentencePiece

Data-driven, language-independent subword segmentation.

SentencePiece segmented Pa’O sentence:
po: __ဝွေ့ __မူး __ခြင် ်း __နှယ် __လို __ပေး __ဖုံး __ဝဝ်း __ခြင် ်း __တ __ပ ှ ဒ် __ ။

SYSTEM COMBINATIONS

Strategic System Combinations Achieve Significant Performance Improvements

The study evaluated multiple combination strategies to leverage complementary strengths of different translation methods, leading to better final translations than single approaches.

Selection Mechanism: Translated sentences from each approach are evaluated and the sentence with the higher **RIBES score** is selected as the final output.

Four Strategic Combinations

COMBINATION 1

PBSMT + HPBSMT

COMBINATION 2

PBSMT + OSM

COMBINATION 3

HPBSMT + OSM

COMPREHENSIVE COMBINATION

PBSMT + HPBSMT + OSM

EVALUATION METRICS

Comprehensive Evaluation Using Three Automatic Assessment Metrics

BLEU (Bilingual Evaluation Understudy)

Measures the precision of n-gram matches (with n ≤ 4) between reference and translated text, with a penalty for short translations. **Large BLEU scores indicate better translation quality.**

RIBES (Rank-based Intuitive Bilingual Evaluation Score)

Based on rank correlation coefficients and provides special consideration for word order in translation evaluation, making it particularly suitable for **distance language pairs** such as Burmese and English.

chrF++ (Character n-gram F-score)

Represents a very promising evaluation metric for machine translation, especially for morphologically rich target languages. This metric operates at the **character level rather than word level**, providing more nuanced evaluation of translation quality.

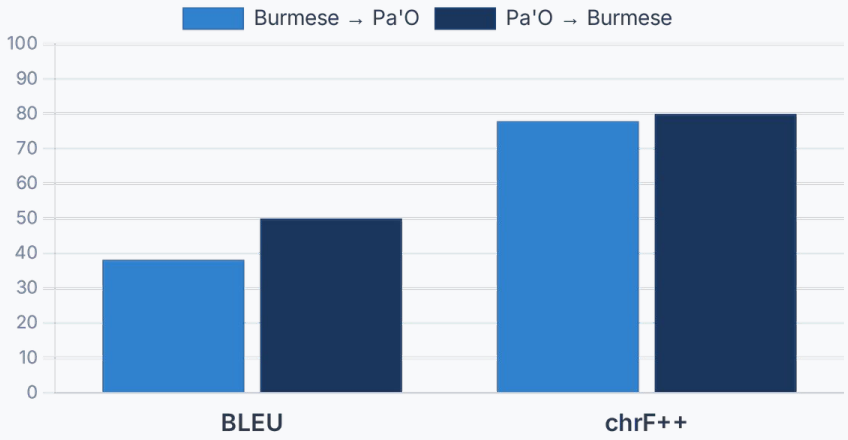
KEY FINDINGS

PBSMT+HPBSMT+OSM Achieves Highest Translation Quality

The combined PBSMT+HPBSMT+OSM model with syllable-SentencePiece segmentation achieved the best results across metrics. Pa'O→Burmese outperforms the reverse direction, largely because OOV issues are less frequent when translating into Burmese.

> ★ **The OSM approach alone achieved the highest BLEU and chrF++ scores among individual methods.**

BLEU and chrF++ Scores (Syllable-SentencePiece)



RIBES (BURMESE → PA'O)	RIBES (PA'O → BURMESE)
0.868276	0.902760

Detailed Performance Metrics with Syllable Segmentation

System Combination	BLEU		RIBES		chrF++	
	bm-po	po-bm	bm-po	po-bm	bm-po	po-bm
PBSMT+HPBSMT	35.55	44.83	0.853441	0.888488	c6+w2-F2:74.9682 c6+w2-avgF2:75.5420	c6+w2-F2:76.2639 c6+w2-avgF2:77.1372
PBSMT+OSM	37.00	45.56	0.851019	0.881752	c6+w2-F2:75.4682 c6+w2-avgF2:75.9877	c6+w2-F2:76.4553 c6+w2-avgF2:77.2992
HPBSMT+OSM	37.54	46.77	0.860096	0.891996	c6+w2-F2:75.4529 c6+w2-avgF2:76.0592	c6+w2-F2:77.0910 c6+w2-avgF2:77.9927
PBSMT+HPBSMT+OSM	**38.45**	**47.57**	**0.868276**	**0.896817**	**c6+w2-F2:75.8660 c6+w2-avgF2:76.4931**	**c6+w2-F2:77.2953 c6+w2-avgF2:78.2121**

Note: TABLE I: BLEU, RIBES and chrF++ scores for system combination using syllable segmentation. Bold numbers indicate the highest scores among system combinations.

RESULTS: SYLLABLE-SENTENCEPIECE

Superior Performance with Syllable-SentencePiece Segmentation

System Combination	BLEU		RIBES		chrF++	
	bm-po	po-bm	bm-po	po-bm	bm-po	po-bm
PBSMT+HPBSMT	36.78	47.40	0.845609	0.891085	c6+w2-F2:77.2030 c6+w2-avgF2:77.8438	c6+w2-F2:79.1240 c6+w2-avgF2:79.9693
PBSMT+OSM	36.92	47.60	0.840138	0.881706	c6+w2-F2:77.2778 c6+w2-avgF2:77.7723	c6+w2-F2:79.0864 c6+w2-avgF2:79.9170
HPBSMT+OSM	37.16	49.29	0.849800	0.896932	c6+w2-F2:77.6732 c6+w2-avgF2:78.3184	c6+w2-F2:79.7598 c6+w2-avgF2:80.6516
PBSMT+HPBSMT+OSM	**38.46**	**50.03**	**0.857091**	**0.902760**	**c6+w2-F2:77.9013** **c6+w2-avgF2:78.5511**	**c6+w2-F2:79.9984** **c6+w2-avgF2:80.9252**

Note: TABLE II: BLEU, RIBES and chrF++ scores for system combination using syllable-SentencePiece segmentation. The **PBSMT+HPBSMT+OSM** combination achieved the highest scores.

ERROR ANALYSIS

Understanding Translation Errors and Confusion Pairs

Translation Error Analysis

Word Error Rate (WER) analysis was conducted using the SCLITE program from the NIST scoring toolkit. The WER formula accounts for insertions, deletions, and substitutions in the translation output to comprehensively evaluate system performance.

Out Of Vocabulary (OOV) words present a significant challenge, particularly in Burmese to Pa'O translation, serving as a primary source of translation errors.

Top 10 Confusion Pairs

Freq	Confusion Pair (REF ==> HYP)
26	တဝ်းဟောင်: ==> ဟောင်:
23	နေ့ဟောင်: ==> ဟောင်:
16	တမ့ွေးတဝ်းဟောင်: ==> မ့ွေးတဝ်းဟောင်:
15	ဒ္ဗား: ==> လ့ွမ်
13	နော: ==> ကရ့ိ
12	ကို ==> နဝ်,
12	ဒ္ဗား: ==> ဟောင်:
12	ယို ==> ယိုနဝ်,
10	ဗား:တဝ်: ==> တဝ်:
10	ဝ့ွေ,နဝ်, ==> ဝ့ွေ,

Table III: illustrates the most frequent confusion pairs in the Operation Sequence Model (OSM) for Burmese-Pa'O translation with word segmentation.

CONCLUSION

Multi-System Approach Opens New Possibilities for Minority Language Translation

Validation of System Combinations

Combining statistical MT approaches improved Burmese–Pa'O translation. The syllable–SentencePiece segmentation performed best, suggesting hybrid segmentation better captures these languages.

Future Research Directions

Explore neural MT and integrate linguistic knowledge to further enhance translation quality for this language pair.

Implications for Pa'O Communities

This system supports Pa'O communities by improving access to information and advancing translation tools for low-resource languages.

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Research Publication

Related Journal Paper

International Journal:

Hay Man Htun, Ye Kyaw Thu, Hlaing Myat Nwe, May Thu Win, Naw Naw, "Statistical Machine Translation System Combinations on Phrase-based, Hierarchical Phrase-based and Operation Sequence Model for Burmese and Pa'O Language Pair", Journal of Intelligent Informatics and Smart Technology, Oct 2nd Issue, 2021, pp. 1-9. (Submitted July 16, 2021; accepted October 13, 2021; revised October 20,2021; published online October 31, 2021)

Paper Link:

https://drive.google.com/file/d/1OAHXKAXM8Rk9yJ6D0AN9gQ9LMoSPsMIs/view?usp=drive_link

Language Understanding λ(μ) Laboratory

11 April 2026

myMediCon: End-to-End Burmese Automatic Speech Recognition for Medical Conversations

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Outline

- Introduction
- Objective
- Burmese
- Building myMediCon Corpus
- Methods and Experimental Setup
- Results and Discussion
- Error Analysis
- Conclusion and Future Work
- References

Introduction

ASR in healthcare: Enables real-time transcription for accurate documentation, allowing medical professionals to focus on patients, not notes.

Accuracy: Ranges from **78% to 92%**, indicating variability (Goss et al., 2019; Sima, 2016).

Simplified Processes: End-to-end (E2E) architectures simplify ASR modeling and decoding processes by incorporating acoustic, pronunciation, and language models into a single neural network.

Data Dependency: E2E model performance heavily relies on the scale of training data, requiring large datasets for optimal results.

Challenges in Burmese ASR: Limited research and scarce speech corpora hinder the development of ASR for Burmese, particularly in specialized domains like medicine.

Objective

1. To develop a publicly available medical domain speech corpus for Burmese ASR.
2. To evaluate corpus using E2E ASR technologies (RNN and Transformer) for accuracy assessment.
3. To compare the results of RNN and Transformer models.
4. To make detail error analysis.

Burmese

- Burmese, Myanmar's official language, spoken by approximately 32 million as a first language and 10 million as a second.
- Burmese text lacks explicit word boundaries and spaces between words.
- It consists of 33 consonants, 12 vowels, and 4 medials.
- Due to limited resources, Burmese is considered under-resourced.

The following is an example of word-level segmented Burmese sentence from medical conversations:

Burmese Sentence:	**ကင်ဆာ ဖြစ် နိုင် လား ၊ ဒေါက်တာ ။**
English Translation:	**Could it be cancer, Doctor?**

The following is an example of syllable-level segmented Burmese sentence from medical conversations:

Burmese Sentence:	**ခင် ဗျား မှာ ဆီး ချို ရော ဂါ ၊ သွေး တိုး ရော ဂါ ကဲ့ သို့ သော ရော ဂါ အ ခြေ အ နေ မျိုး တွေ ရှိ ပါ သ လား ။**
English Translation:	**Do you have any medical problems like diabetes mellitus, high blood pressure?**

Building myMediCon Corpus

- Main Contribution: Created "**myMediCon**," a corpus of Burmese Medical Speech Conversations.
- Constructed by gathering Burmese medical text and recording speech data via natural reading of collected sentences.
- Medical sentences sourced from "Samson Handbook of Plab 2 and Clinical Assessment" (Samson, 2015), manually translated from English to Burmese, covering patient-doctor conversations, disease and medicine names, and treatment methodologies.

Medical text corpus:
- Sentences: 14,592
- Total words: 232,999
- Unique words: 15,431

Recording devices:
- TASCAM (DR44-WL) recorder
- Laptop (Dell i7 8th Gen)
- Smartphones (iPhone, Oppo, Redmi, Xiao Mi)

Speakers:
- Total: 12
- Gender: 3 males, 9 females
- Age range: 20 to 30 years

Cumulative audio duration:
- Total: 93 hours, 59 minutes, 43 seconds

Individual utterance duration:
- Range: 1.8 to 35 seconds

Cont'd

Speaker ID	Utterance	Duration
spk00	1001	2 hr, 1 min, 41 sec
spk01	14592	26 hr, 16 min, 35 sec
spk02	14592	37 hr, 33 min, 5 sec
spk03	1000	2 hr, 56 min, 28 sec
spk04	1000	1 hr, 57 min, 40 sec
spk05	2000	3 hr, 52 min, 24 sec
spk06	1500	3 hr, 5 min, 21 sec
spk07	1000	1 hr, 48 min, 23 sec
spk08	500	59 min, 3 sec
spk09	500	46 min, 55 sec
spk10	4092	7 hr, 53 min, 9 sec
spk11	3000	4 hr, 48 min, 59 sec

Table 1: Total duration and utterances of each speaker in **myMediCon** corpus.

Methods and Experimental Setup

- We utilized ESPnet[1], a PyTorch-based speech processing toolkit, to develop and evaluate end-to-end ASR models (RNN and Transformer) on **myMedicon,** our medical speech corpora.

RNN (Recurrent Neural Network)

RNNs in ASR:

- Widely used for flexibility and temporal dependency capture (Suyanto et al., 2020; Schmidt, 2019).

Encoder:

- VGG-RNN design with four bidirectional LSTM layers for temporal pattern capture.

Decoder:

- Unidirectional LSTM layer refines representations for improved accuracy.

Training:

- Robust scheduling mechanism and Adadelta optimizer with dynamic learning rate adjustment (ReduceLROnPlateau scheduler with patience set to 1 and mode set to minimize, learning rate of 1.0).

[1]Available at https://github.com/espnet/espnet

Cont'd

Transformer

- **Transformer architecture:** Replaces RNNs and utilizes self-attention mechanism for NLP tasks (Karita et al., 2019).
- **Encoder:** 12 blocks, each with 4 attention heads, utilizing attention and positional dropout techniques.
- **Decoder:** Three transformer blocks for efficient context integration, incorporating label smoothing (weight: 0.1) and joint CTC-attention training (weight: 0.3).
- **Training:** Warmup learning rate scheduler and Adam optimizer with learning rate of 0.001.

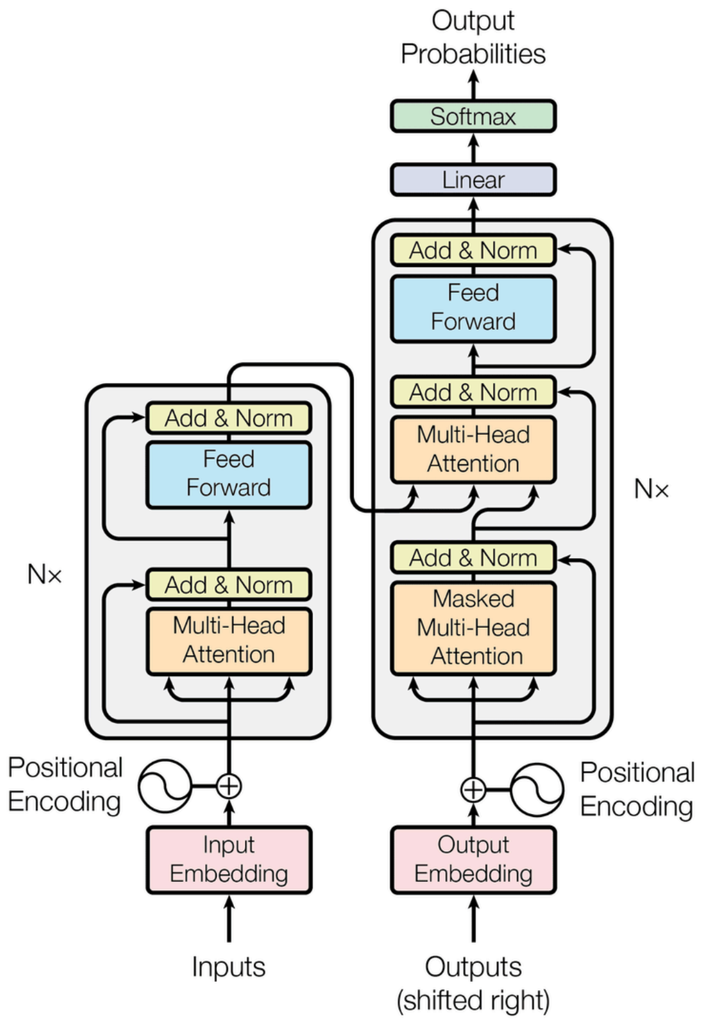

Figure 1: Transformer model architecture (Vaswani et al., 2017)

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Evaluation Metrics

- **Word Error Rate** (WER), **Character Error Rate** (CER), and **Translation Error Rate** (TER) are common metrics for evaluating ASR, Optical Character Recognition (OCR), and Machine Translation (MT) systems.
- WER assesses word-level errors, while CER does so at the character level. TER extends to MT, including shifts in text.
- Lower values mean better accuracy, offering quantitative measures of quality.

Data Statistics

Burmese speech and text corpus for experiments: • Total sentences: 5,400 • Sentences per speaker: 450 • Selected randomly from myMediCon corpus • Total words: 82,218 • Unique words: 7,212	**Total audio duration:** • 10 hours, 46 minutes, 1.35 seconds • Utterance duration range: 1.8 to 35 seconds **Speakers:** • Total: 12 • Gender: 3 males, 9 females

Cont'd

Speaker ID	Utterance	Duration
spk00	450	55 min, 50 sec
spk01	450	39 min, 57 sec
spk02	450	1 hr, 16 min, 18 sec
spk03	450	1 hr, 25 min, 51 sec
spk04	450	55 min, 44 sec
spk05	450	50 min, 30 sec
spk06	450	53 min, 46 sec
spk07	450	38 min, 23 sec
spk08	450	52 min, 57 sec
spk09	450	42 min, 45 sec
spk10	450	46 min, 53 sec
spk11	450	47 min, 02 sec

Table 2: Total duration and utterances of each speaker in the **extracted Burmese Speech corpus** for the system.

Cont'd

Three ASR experiments were conducted using random sub-sampling validation.

- **Experiment 1:** Training set - 11 speakers (2 males, 9 females); Development set - 2 speakers (1 male, 1 female); Test set - 12 speakers (3 males, 9 females).
- **Experiment 2:** Training set - 11 speakers (3 males, 8 females); Development set - 2 speakers (both females); Test set - 12 speakers (3 males, 9 females).
- **Experiment 3:** Equal distribution of 12 speakers (3 males, 9 females) in each set.

Datasets	Utterance	Duration
Train	4200	8 hr, 23 min, 1 sec
Dev	600	1 hr, 9 min, 45 sec
Test	600	1 hr, 13 min, 14 sec

Table 3: Statistics on **train, development and test** sets used for the **1st experiment** of the system.

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Datasets	Utterance	Duration
Train	4200	7 hr, 50 min, 28 sec
Dev	600	1 hr, 43 min, 27 sec
Test	600	1 hr, 12 min, 12 sec

Table 4: Statistics on **train, development and test** sets used for the **2nd experiment** of the system.

Datasets	Utterance	Duration
Train	4200	8 hr, 25 min, 43 sec
Dev	600	1 hr, 9 min, 45 sec
Test	600	1 hr, 9 min, 7 sec

Table 5: Statistics on **train, development and test** sets used for the **3rd experiment** of the system.

Results and Discussion

- Word-level and syllable-level segmentation used on the text corpus.
- A syllable segmentation tool named “sylbreak” was utilized to convert Burmese words to syllable levels.[2]
- Average evaluation results of three experiments with RNN and Transformer models presented in Table 6 and Table 7.

Corpus Types	WER%	CER%	TER%
Word	40.2	**13.6**	20.7
Syllable	25.8	14.1	20.9

Table 6: **Average** evaluation results of **RNN** model for E2E Burmese ASR with WER, CER and TER of the three experiments.

Corpus Types	WER%	CER%	TER%
Word	53.4	18.5	27.3
Syllable	26.4	**12.7**	19.5

Table 7: **Average** evaluation results of **Transformer** model for E2E Burmese ASR with WER, CER and TER of the three experiments.

[2] Available at https://github.com/ye-kyaw-thu/sylbreak

Cont'd

- Best evaluation results of three experiments with RNN and Transformer models are displayed in Tables 8 and 9.
- Syllable-level segmentation significantly improves RNN and Transformer model accuracy.
- RNN model outperforms Transformer across all metrics and segmentation levels.

Corpus Types	WER%	CER%	TER%
Word	36.4	11.5	17.3
Syllable	19.0	**9.7**	14.7

Table 8: The **best** evaluation results of **RNN** model for E2E Burmese ASR with with WER, CER and TER among the three experiments.

Corpus Types	WER%	CER%	TER%
Word	50.7	17.3	25.5
Syllable	23.1	**11.2**	17.1

Table 9: The **best** evaluation results of **Transformer** model for Burmese ASR with WER, CER and TER among the three experiments.

Error Analysis

No.	Frequency	REF ⇒ HYP
1:	26	မ္ ⇒ မ္ာ
2:	17	ရ ⇒ ရာ
3:	12	တ ⇒ တစ်
4:	9	င့် ⇒ နဲ့
5:	9	င့် ⇒ မြင်
6:	9	သော ⇒ တော့
7:	8	စ ⇒ စား
8:	8	စာ ⇒ စား
9:	8	တဲ့ ⇒ နဲ့
10:	8	တွင်း ⇒ တွင်

Table 10: Top 10 errors of **RNN ASR model** with **syllable segmentation**

No.	Frequency	REF ⇒ HYP
1:	10	မ္ ⇒ မ္ာ
2:	9	တာလဲ ⇒ လဲ
3:	9	တဲ့ ⇒ နဲ့
4:	9	နှင့် ⇒ နဲ့
5:	7	အလေးချိန် ⇒ ကိုယ်အလေးချိန်
6:	7	ကံမကောင်းစွာ ⇒ စွာ
7:	7	ဖြစ်စေ ⇒ စေ
8:	6	မ္ာ ⇒ ထဲမ္ာ
9:	6	တာကို ⇒ ကို
10:	8	ထွက် ⇒ သွေးထွက်

Table 11: Top 10 errors of **RNN ASR model** with **word segmentation**

Cont’d

- **SCLITE** highlighted top 10 errors from RNN models, displayed in Tables 10 and 11, mainly due to phonetic discrepancies, text encoding, and word segmentation challenges.
- Phonetic errors included instances like "နှင့်" pronounced as "nhin." and "ရှိ" as "ne."
- Errors related to incorrect typing order of Burmese syllables, e.g., "င့်" and "ရှိ".
- Mitigation involves expanding speech corpus and normalizing text transcriptions.

Conclusion and Future Work

- Introduced low-resourced **Burmese ASR** in the medical domain using RNN and Transformer architectures within ESPnet framework.
- Developed "**myMediCon**," a Burmese Medical Speech Conversations corpus.
- Syllable-level segmentation outperformed word-level, with **RNN** achieving the lowest CER of **9.7%**.
- Plan to publish myMediCon corpus for future Burmese speech researchers.
- Future work includes extending E2E Burmese ASR experiments with advanced techniques and an expanded Burmese speech corpus in the medical domain.

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Research Publication of the Proposed ASR System

- **Paper Title:** Hay Man Htun, Ye Kyaw Thu, Hutchatai Chanlekha, Kotaro Funakoshi and Thepchai Supnithi, "myMedicon: End-to-End Burmese Automatic Speech Recognition for Medical Conversations", the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), 20-25 May, Torino, Italy, pp. 12032–12039
- **Paper Link:** https://aclanthology.org/2024.lrec-main.1051/

Thank You For Your Attention

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